Foundations of Full House Methodology

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Abstract

The Full House Methodology is an approach for computing era-adjusted or system-adjusted statistics that balances the size of the source population in each system with the component's performance via "versus their peers" in a specific system. We assume that each component has a latent aptitude that can be calculated from this balancing act provided the system inclusion mechanism for components of the system is known. Without making any assumptions about the distribution of the system's components, the components' distribution is well-established from the nonparametric probability distribution.

1 Introduction

Several academics have attempted to develop methods that can correct for era or system biases. The starting place for the answer to the problem of comparing players across eras can be found in the work of Stephen Jay Gould. Gould [1996] suggested that the entire distribution of achievement, or "full house of variation," is relevant when considering the achievements of baseball players across eras. This idea was also recognized as the starting place for an answer by Schell [2013] and Schell [2016].

Berry et al. [1999], Petersen et al. [2011], Schell [2013, 2016], Petersen and Penner [2020] provide several techniques to compare baseball players across eras, but they have fallen short in their objective and their results remain biased, as discussed in Eck [2020a] and Eck [2020b]. In short, these previous attempts do not incorporate the evolving talent pool in calculations, which is necessary for understanding the changing distributions of achievement over time. The result is that existing methodologies produce a disproportionate number of players who began their career before baseball was integrated players in the rankings of baseball's greatest of all time. This era of baseball did not include a crucial population of eligible players. Yu et al. [2016], Meyer [2021], Campbell [1990] also develop methods for comparing components in historical popularity, educational attainment, and evaluation of employees. But they have similar problems as stated before and fail to account for the evolving talent pool.

In some circumstances, such as test equating, it is not required to take into account an evolving talent pool. Andersson et al. [2013] offers a collaborative framework for test equating and builds the cumulative density function using the kernel method. Nevertheless,

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they are unable to resolve the extrapolation issue in non-parametric estimation, and the distribution's tail is not given special consideration.

Our Full House methodology directly incorporates the quality of the talent pool as a central modeling component. It offers metrics for assessing the performance of components in various eras or systems. With the application of educational attainment simulation, our Full House methodology reveals that the most talented students could perform worse than the other students who attend schools with lower mean scores.

The next section establishes the model setting and specifies the assumptions in our Full House methodology. Section 3 considers an intriguing paradox that the component with more talent underperforms the component with less talent. We compute the probability of this paradox under different scenarios. Section 4 discusses the application of the Full House methodology for the educational attainment simulation.

2 Model Setting

We now motivate the structure of the Full House Model. Let N_i denote the size of population i. We will suppose the every individual $k = 1, \ldots, N_i$ has an underlying latent $\mathbb{R}$-valued aptitude $X_{i,1}, \ldots, X_{i,N_i} \overset{iid}{\sim} F_X$. Let $\mathbf{X}_{i,-j}$ be the vector of all latent aptitudes not including the component j. We then define an indicator function $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right)$ as a determined and known system inclusion function for population i where a 1 indicates inclusion and a 0 indicates exclusion. We now suppose that an individual j included in system i has an observed $\mathbb{R}$-valued statistic $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$, $n_i \leq N_i$. As we will see later, the observed $Y_{i,j}$ values are one-to-one realizations of $X_{i,j}$, provided that $g_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1$. Therefore $Y_{i,1}, \ldots, Y_{i,n_i} \overset{iid}{\sim} F_{Y_i}$.

Let $X_{i,(j)}$ and $Y_{i,(j)}$ denote the jth order statistic for system i. Here, larger numbers will correspond to a notion of "better." Thus $X_{i,(j)} > X_{i,(k)}$ whenever $j > k$, for example. We now form (Y, X) pairs of random variables for each individual included in the system. These pairs are of the form $(Y_{i,(j)}, X_{i,(r_{i,j})})$ where $r_{i,j}$ indexes the aptitude of the jth highest value observed from the system. As an example, let $1_i\left(X_{i,j}, \mathbf{X}_{i,-j}\right) = 1(X_{i,j} \geq X_{i,(N_i - n_i + 1)})$, then the $(Y_{i,(j)}, X_{i,(r_{i,j})})$ pairs take the form $(Y_{i,(j)}, X_{i,(N_i - n_i + j)})$ for $j = 1, \ldots, n_i$.

The key Full House Modeling is the explicit connection between the elements forming the $(Y_{i,(j)}, X_{i,(r_{i,j})})$ pairs. Full House Modeling states this connection as

$$\begin{aligned} Y_{i,(j)} &= F^{-1}_{Y_{i,(j)}} \circ F_{X_{i,(r_{i,j})}}\left(X_{i,(r_{i,j})}\right), \\ X_{i,(r_{i,j})} &= F^{-1}_{X_{i,(r_{i,j})}} \circ F_{Y_{i,(j)}}\left(Y_{i,(j)}\right), \end{aligned} \tag{1}$$

where $1_i(X_{i,(r_{i,j})}, \mathbf{X}_{i,-(r_{i,j})}) = 1$, and both F_{Y_i} and F_X are continuous and invertible. We summarise Full House Modeling in the definition below.

Definition 2.1 (Full House Model)**.** *Consider populations with sizes N_i and system sizes n_i, $N_i \geq n_i$. Let a system inclusion function $1_i(\cdot, \cdot)$ be specified. Let $X_{i,j} \overset{iid}{\sim} F_X$, $j = 1, \ldots, N_i$. Let $Y_{i,j} \overset{iid}{\sim} F_{Y_i}$, $j = 1, \ldots, n_i$. Form pairs $(Y_{i,(j)}, X_{i,(r_{i,j})})$, $j = 1, \ldots, n_i$ for all individuals such that $1_i(X_{i,(r_{i,j})}, \boldsymbol{X}_{i,-(r_{i,j})}) = 1$. Then the Full House Model generates $Y_{i,(j)}$ values following (1) where F_{Y_i} and F_X are continuous and invertible.*

Note that Definition 2.1 fixes F_X but allows F_{Y_i} to vary across systems.

Remarks:

1. The definition of a Full House Model immediately implies structural missingness. Every $X_{i,j}$ such that $1_i(X_{i,j}, \mathbf{X}_{i,-j}) = 0$ will not have a corresponding Y value. In the following sections we will consider probabilities involving both X and Y quantities. To handle this structural missingness, we will suppose that $Y_{i,j}$ values that are not observed are not P-measurable.

3 Paradoxes Under Identical Distributions

In this section we will provide explicit paradox probability results in a Full House Modeling setting involving two populations with the additional specification that $F_{Y_i} = F_Y$, $i = 1, 2$ in Definition 2.1. The following definition formally defines the notion of paradox used throughout this article.

Definition 3.1. *Consider the Full House Model in Definition 2.1 with two populations denoted by $i = 1, 2$. We say a paradox has occurred when both $X_{1,(r_{1,j})} > X_{2,(r_{2,k})}$ and $Y_{1,(j)} < Y_{2,(k)}$ have occurred. The paradox probability is defined as*

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right). \tag{2}$$

As a simple illustrative example of the above, let $r_{i,t} = N_i - n_i + t$ and set $j = n_1$ and $k = n_2$, and further suppose that $Y_{1,j} < Y_{2,k}$ a.s. Then the paradox probability (2) is equivalent to $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right)$, and we have $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right) \to 1$ when $N_1 \to \infty$. We now introduce a convenient representation which allows the paradox probability to be expressed in terms of the latent aptitudes alone.

Lemma 3.2. *Consider the Full House Model in Definition 2.1 with two populations, where $F_{Y_i} = F_Y$ for $i = 1, 2$. Then,*

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right), \tag{3}$$

where $h_{1,2,j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$.

Proof. Define $f_{1,j}(\cdot) = F^{-1}_{Y_{1,(j)}} \circ F_{X_{1,(r_{1,j})}}(\cdot)$ and $f_{2,k}(\cdot) = F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. Then we have

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&\quad = \mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}), f_{1,j}(X_{1,(r_{1,j})}) < f_{2,k}(X_{2,(r_{2,k})})\right) \\
&\quad = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right),
\end{aligned}$$

where $h_{1,2,j,k}(\cdot) = (f^{-1}_{1,j} \circ f_{2,k})(\cdot)$. □

Our first formal result provides conditions that allow for an explicit paradox probability result. Those conditions require the following definition.

Definition 3.3. *Let $\alpha, \beta > 0$. Then the regularized incomplete beta function is defined as*

$$I_{\alpha,\beta}(x) = \frac{1}{B(\alpha,\beta)} \int_0^x u^{\alpha-1}(1-u)^{\beta-1} du,$$

where $B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the beta coefficient.

Note that the regularized incomplete beta function is the cumulative distribution function of a Beta(α,β) distribution.

Theorem 3.4. *Consider the Full House Model in Definition 2.1 with two populations, where $F_{Y_i} = F_Y$ for $i = 1, 2$. Denote by I the regularized incomplete beta function. Further assume that*

$$I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \geq 0, \qquad \text{for all } t \in (0,1). \tag{4}$$

Under these assumptions,

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) \\
&= \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2-n_1-p+k+1)}{B(k, n_2-k+1)} - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k}+p, N_2-N_1-p+r_{2,k}+1)}{B(r_{2,k}, N_2-r_{2,k}+1)}.
\end{aligned}$$

Remarks:

1. The result of Theorem 3.4 is distribution free. It only depends on the sample sizes n_i, N_i, $i = 1, 2$, the system inclusion function, and the indexes of the order statistics under comparison. The distribution on Y is eliminated by Lemma 3.2. To see this, consider the distribution $F_{Y_{(j)}}$ for some $j = 1, \ldots, n_1$. The random variable $Y_{(j)} = F_Y^{-1} \circ U_{(j)}$ where $U_{(j)} \sim \text{Beta}(j, n_1 - j + 1)$. Thus, $F_{Y_{(j)}}(y) = I_{j,n_1-j+1} \circ F_Y(y)$. With this in mind, from Lemma 3.2 we have that

$$\begin{aligned}
h_{1,2,j,k}(x) &= F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(x) \\
&= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ \left(F_Y \circ F_Y^{-1}\right) \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x) \\
&= F^{-1}_{X_{1,(r_{1,j})}} \circ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1} \circ F_{X_{2,(r_{2,k})}}(x).
\end{aligned}$$

   The above is the upper-bound of an integral with respect to Xs in the proof of Theorem 3.4. The influence of the distributional specification on Y does not influence the calculation of the paradox probability result in Theorem 3.4.

2. The distribution on X is eliminated when calculating the paradox probability (2) in its Lemma 3.2 representation, $\mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right)$. These calculations involve double integrals with respect to densities for $F_{X_{1,(r_{1,j})}}$ and $F_{X_{2,(r_{2,k})}}$. One integral evaluation and a reparameterization remove the specification of F_X from the probability calculation as seen in the proof of Theorem 3.4.

3. The steps mentioned in the last sentence of the previous remark allow the probability calculation for $\mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2}\left(X_{2,(r_{2,k})}\right)\right)$ to simplify to

$$\int_0^1 \left\{ I_{j,n_1-j+1} \circ I^{-1}_{k,n_2-k+1}(t) - I_{r_{1,j},N_1-r_{1,j}+1} \circ I^{-1}_{r_{2,k},N_2-r_{2,k}+1}(t) \right\}^+ dt.$$

The technical condition (4) ensures that this integral defines a non-negative probability in strong terms. We demonstrate with analytic and numeric arguments when (4) is satisfied and the results of Theorem 3.4 therefore hold.

Proof. First, Lemma 3.2 allows us to write

$$\mathrm{P}\left(X_{1,(r_{1,j})} > X_{2,(r_{2,k})}, Y_{1,(j)} < Y_{2,(k)}\right) = \mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2,j,k}\left(X_{2,(r_{2,k})}\right)\right),$$

where $h_{1,2,j,k}(\cdot) = F^{-1}_{X_{1,(r_{1,j})}} \circ F_{Y_{1,(j)}} \circ F^{-1}_{Y_{2,(k)}} \circ F_{X_{2,(r_{2,k})}}(\cdot)$. We can rewrite $h_{1,2,j,k}(\cdot)$ into a form that will allow for specific computation involving Beta distributions. o see this, consider the distribution $F_{Y_{(j)}}$ for some $j = 1, \ldots, n_1$. The random variable $Y_{(j)} = F^{-1}_Y \circ U_{(j)}$ where $U_{(j)} \sim \mathrm{Beta}(j, n_1 - j + 1)$. Thus, $F_{Y_{(j)}}(y) = g_{n,j} \circ F_Y(y)$, where we write $g_{n,j}(x) = I_{j,n-j+1}(x)$ for ease of notation. This expression of distribution functions allows us to write

$$\begin{aligned}
&\mathrm{P}\left(X_{2,(r_{2,k})} < X_{1,(r_{1,j})} < h_{1,2,j,k}\left(X_{2,(r_{2,k})}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left\{ \int_x^{F_X^{-1} \circ g^{-1}_{N_1,r_{1,j}} \circ g_{n_1,j} \circ g^{-1}_{n_2,k} \circ g_{N_2,r_{2,k}} \circ F_X(x)} f_{X_{1,(r_{1,j})}}(y)\, dy \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ \int_x^{F_X^{-1} \circ g^{-1}_{N_1,r_{1,j}} \circ g_{n_1,j} \circ g^{-1}_{n_2,k} \circ g_{N_2,r_{2,k}} \circ F_X(x)} d\, g_{N_1,r_{1,j}}(F_X(y)) \right\} f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ g_{n_1,j} \circ g^{-1}_{n_2,k} \circ g_{N_2,r_{2,k}} \circ F_X(x) - g_{N_1,r_{1,j}} \circ F_X(x) \right\}^+ f_{X_{2,(r_{2,k})}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left\{ g_{n_1,j} \circ g^{-1}_{n_2,k} \circ g_{N_2,r_{2,k}} \circ F_X(x) - g_{N_1,r_{1,j}} \circ F_X(x) \right\}^+ d\, g_{N_2,r_{2,k}}(F_X(x)) \\
&= \int_0^1 \left\{ g_{n_1,j} \circ g^{-1}_{n_2,k}(t) - g_{N_1,r_{1,j}} \circ g^{-1}_{N_2,r_{2,k}}(t) \right\}^+ dt.
\end{aligned}$$

Condition (4) states that

$$\int_0^1 \left\{ g_{n_1,j} \circ g^{-1}_{n_2,k}(t) - g_{N_1,r_{1,j}} \circ g^{-1}_{N_2,r_{2,k}}(t) \right\}^+ dt = \int_0^1 \left\{ g_{n_1,j} \circ g^{-1}_{n_2,k}(t) - g_{N_1,r_{1,j}} \circ g^{-1}_{N_2,r_{2,k}}(t) \right\} dt.$$

Thus,

$$\begin{aligned}
&\int_0^1 \left\{ g_{n_1,j} \circ g^{-1}_{n_2,k}(t) - g_{N_1,r_{1,j}} \circ g^{-1}_{N_2,r_{2,k}}(t) \right\}^+ dt \\
&\quad = \int_0^1 g_{n_1,j}(x)\, dg_{n_2,k}(x) - \int_0^1 g_{N_1,r_{1,j}}(x)\, dg_{N_2,r_{2,k}}(x) \\
&\quad = \int_0^1 I_x(j, n_1 - j + 1) \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2 - k + 1)}\, dx
\end{aligned}$$

$$
\begin{aligned}
&\quad - \int_0^1 I_x(r_{1,j}, N_1 - r_{1,j} + 1) \frac{x^{r_{2,k}-1}(1-x)^{N_2 - r_{2,k}}}{B(k, N_2 - k + 1)}\, dx \\
&= \int_0^1 \sum_{p=j}^{n_1} \binom{n_1}{p} x^p (1-x)^{n_1 - p} \frac{x^{k-1}(1-x)^{n_2-k}}{B(k, n_2 - k + 1)}\, dx \\
&\quad - \int_0^1 \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} x^p (1-x)^{N_1 - p} \frac{x^{r_{2,k}-1}(1-x)^{N_2 - r_{2,k}}}{B(r_{2,k}, N_2 - r_{2,k} + 1)}\, dx \\
&= \sum_{p=j}^{n_1} \binom{n_1}{p} \frac{B(k+p, n_2 + n_1 - p - k + 1)}{B(j, n_2 - k + 1)} \\
&\quad - \sum_{p=r_{1,j}}^{N_1} \binom{N_1}{p} \frac{B(r_{2,k} + p, N_2 + N_1 - p - r_{2,k} + 1)}{B(r_{2,k}, N_2 - r_{2,k} + 1)}.
\end{aligned}
$$

□

Often it is of interest to compare the individuals with the highest aptitudes in their respective populations. Below we state the paradox probability in such setting using a Full House Model with two populations with the specification that $F_{Y_1} = F_Y$ for $i = 1, 2$.

Corollary 3.4.1. *Suppose that the conditions of Theorem 3.4 hold. Suppose that* $1_1(X_{1,(N_1)}, \boldsymbol{X}_{1,-(N_1)}) = 1_2(X_{2,(N_2)}, \boldsymbol{X}_{2,-(N_2)}) = 1$. *Then,*

$$
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \begin{cases} \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2}, & \text{if } \frac{n_1}{N_1} \le \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.
$$

A similar result holds for the individuals with the lowest latent aptitudes in the population.

Corollary 3.4.2. *Suppose that the conditions of Theorem 3.4 hold. Suppose that* $1_1(X_{1,(1)}, \boldsymbol{X}_{1,-(1)}) = 1_2(X_{2,(1)}, \boldsymbol{X}_{2,-(1)}) = 1$. *Then,*

$$
\mathrm{P}\left(X_{1,(1)} > X_{2,(1)}, Y_{1,(1)} < Y_{2,(1)}\right) = \begin{cases} \frac{N_2}{N_1+N_2} - \frac{n_2}{n_1+n_2}, & \text{if } \frac{n_1}{N_1} \ge \frac{n_2}{N_2} \\ 0, & \text{otherwise} \end{cases}.
$$

We now demonstrate a setting in which technical condition (4) holds. This setting is based on a configuration of n_j, N_j, $j = 1, 2$ under the assumption that the systems are populated with individuals from the source populations who have the highest aptitudes.

Proposition 3.5. *Consider the Full House Model in Definition 2.1 with two populations, where* $F_{Y_i} = F_Y$ *for* $i = 1, 2$, *and let the system inclusion function be* $1_i(X_{i,j} \ge X_{i,(N_i - n_i + 1)})$. *Denote by* I *the regularized incomplete beta function. Suppose that* $\frac{n_1}{N_1} \le \frac{n_2}{N_2}$ *and set* $k = j$. *Under these assumptions, we conclude that technical condition (4) holds and the conclusion of Theorem 3.4 applies.*

Proof. Specification of $1_i(X_{i,j} \geq X_{i,(N_i-n_i+1)})$ implies that $r_{i,j} = N_i + 1 - j$.

When either $n_1 = N_1$ and $n_2 = N_2$ or $n_1 = n_2$ and $N_1 = N_2$ we have, for all $x \in (0,1)$, that

$$I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t.$$

The left hand side of the above equation increases whenever n_1 or N_1 are increased [Askitis, 2021], or n_2 or N_1 are decreased. Therefore, $I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t$ whenever $\frac{n_1}{n_2} \leq 1$ and $\frac{N_2}{N_1} \leq 1$. Observe that

$$\begin{aligned} & I_{N_2-j,j+1} \circ I^{-1}_{N_1-j,j+1} \circ I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq t, \\ \implies & I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(t) \geq I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t), \\ \implies & I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(t) \geq t. \end{aligned}$$

Technical condition (4) holds and the conclusion of Theorem 3.4 applies. □

4 Paradox Under Different Distribution

Definition 4.1. *We will say that Y_2 is first order ε-weakly stochastically dominant over Y_1 if, for $\varepsilon > 0$, $\mathrm{P}(Y_2 \leq y) \leq \mathrm{P}(Y_1 \leq y) \leq \mathrm{P}(Y_2 \leq y) + \varepsilon$ for all y and $\mathrm{P}(Y_2 \leq y) < \mathrm{P}(Y_1 \leq y) \leq \mathrm{P}(Y_2 \leq y) + \varepsilon$ for some y.*

Proposition 4.2. *Suppose that the system inclusion mechanism has the property that $g(X_{1,(r_{1,k})}) = 1$ and $g(X_{2,(r_{2,j})}) = 1$, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_{Y_i}(y), \qquad i = 1,2, \qquad j = 1,\ldots,n_i, \\ X_{i,j} &\sim F_X(x), \qquad i = 1,2, \qquad j = 1,\ldots,N_i, \end{aligned}$$

where $F^{-1}_{Y_i}(y)$ and $F^{-1}_X(x)$ exist. Further suppose that Y_2 is first order ε-weakly stochastically dominant over Y_1 for some $\varepsilon > 0$. Then,

$$\begin{aligned} \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2} &\leq \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\leq n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2} \end{aligned} \tag{5}$$

Proof. First, observe that the assumption of first order ε-weakly stochastically dominant gives

$$\left(F_{Y_{2,(n_2)}}(y)\right)^{n_1/n_2} \leq \left(F_{Y_1}(y)\right)^{n_1} \leq \left(F_{Y_2}(y)+\varepsilon\right)^{n_1} = \left(\left(F_{Y_{2,(n_2)}}(y)\right)^{1/n_2} + \varepsilon\right)^{n_1},$$

and this implies that

$$F^{-1}_X\left(F_X^{\frac{N_2}{N_1}\frac{n_1}{n_2}}(X_{2,(N_2)})\right) \leq h_{1,2}(X_{2,(N_2)}) \leq F^{-1}_{X_{1,(N_1)}}\left(\left(\left(F_{X_{2,(N_2)}}(X_{2,(N_2)})\right)^{1/n_2} + \varepsilon\right)^{n_1}\right).$$

In combination with Corollary 3.4.1 and the proof of Corollary 3.4.1, the first inequality yields

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \geq \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2}.$$

This verifies the first inequality of (5). We now verify the second inequality of (5). Now,

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&\quad \leq \int_{-\infty}^{\infty} \left[\int_{x}^{F^{-1}_{X_{1,(N_1)}}\left(\left(\left(F_{X_{2,(N_2)}}(x)\right)^{1/n_2}+\varepsilon\right)^{n_1}\right)} f_{X_{1,(N_1)}}(y)dy \right] f_{X_{2,(N_2)}}(x)dx \\
&\quad = \int_{-\infty}^{\infty} \left[\left(\left(F_X(x)\right)^{N_2/n_2} + \varepsilon \right)^{n_1} - F_X(x)^{N_1} \right] N_2 F_X(x)^{N_2-1} f_X(x)dx \\
&\quad = \int_{-\infty}^{\infty} \left(\left(F_X(x)\right)^{N_2/n_2} + \varepsilon \right)^{n_1} N_2 F_X(x)^{N_2-1} f_X(x)dx - \frac{N_2}{N_1+N_2} \\
&\quad = \int_0^1 \left(t^{1/n_2}+\varepsilon\right)^{n_1} dt - \frac{N_2}{N_1+N_2} \\
&\quad = n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2}
\end{aligned}$$

□

Example: Uniform distributions. In this example we demonstrate a case where Prop 4.2 applies, but we can explicitly calculate the paradox probability. Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1$, and we will consider the following Full House Model for two systems,

$$\begin{aligned}
Y_{1,j} &\sim U(0,1), \qquad j = 1,\ldots,n_1, \\
Y_{2,j} &\sim U(0,1+\delta), \qquad j = 1,\ldots,n_2, \\
X_{i,j} &\sim F_X(x), \qquad j = 1,\ldots,N_i,
\end{aligned}$$

where $\delta > 0$. We can pick $\varepsilon = \delta/(1+\delta)$ so that Y_2 is first order ε-weakly stochastically dominant over Y_1 with this choice of ε, and the conditions of Prop 4.2 are satisfied. However, our setup allows for direct calculation of the paradox probability. From Lemma 3.2 and a bit of manipulation, we have that

Theorem 4.3.

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \begin{cases} \frac{n_2}{n_1+n_2}(1+\delta)^{n_1} - \frac{N_2}{N_1+N_2}, & \text{if } \frac{n_1}{N_1} \geq \frac{n_2}{N_2} \\ \frac{N_1}{N_1+N_2} - \frac{n_1}{n_1+n_2}\left(\frac{1}{(1+\delta)}\right)^{n_2} - & \\ \frac{n_2}{n_1+n_2}(1+\delta)^{n_1} d^{\frac{(n_1+n_2)N_2}{n_2}} + \frac{N_2}{N_1+N_2} d^{(N_1+N_2)}, & \text{otherwise} \end{cases}.$$

Proof.

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right)$$

$$
\begin{aligned}
&= \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}, Y_{2,(n_2)} \in (0,1)\right) \\
&+ \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}, Y_{2,(n_2)} \in (1,1+\delta)\right) \\
&= \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < f_{1,2}(X_{2,(N_2)}), F_X(X_2)^{\frac{N_2}{n_2}}(1+\delta) \in (0,1)\right) \\
&+ \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, F_X(X_2)^{\frac{N_2}{n_2}}(1+\delta) \in (1,1+\delta)\right) \\
&= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_{x}^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\ d(F_X(x)^{N_2}) \\
&+ \int_{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})}^{+\infty} \int_{x}^{+\infty} d(F_X(y)^{N_1})\ d(F_X(x)^{N_2}) \\
&= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_{x}^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}})
\end{aligned}
$$

The support for the outer integration above is $\{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}}) > x\} \cup \{x < F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})\}$.

Therefore, when $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$,

$$
\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_{x}^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{-\infty}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} (F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{n_1} - F_X(x)^{N_1}\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{0}^{(\frac{1}{1+\delta})^{n_2}} (1+\delta)^{n_1} t^{\frac{n_1}{n_2}} - t^{\frac{N_1}{N_2}}\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \frac{N_1}{N_1+N_2} - (\frac{1}{1+\delta})^{n_2}\frac{n_1}{n_1+n_2}
\end{aligned}
$$

Therefore, when $\frac{n_1}{N_1} \geq \frac{n_2}{N_2}$, and let $d = (1+\delta)^{\frac{1}{\frac{N_1}{n_1}-\frac{N_2}{n_2}}}$

$$
\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right)
$$

$$
\begin{aligned}
&= \int_{F_X^{-1}(d)}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} \int_x^{F_X^{-1}((F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{\frac{n_1}{N_1}})} d(F_X(y)^{N_1})\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{F_X^{-1}(d)}^{F_X^{-1}((\frac{1}{1+\delta})^{\frac{n_2}{N_2}})} (F_X(x)^{\frac{N_2}{n_2}}(1+\delta))^{n_1} - F_X(x)^{N_1}\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \int_{d^{N_2}}^{(\frac{1}{1+\delta})^{n_2}} (1+\delta)^{n_1} t^{\frac{n_1}{n_2}} - t^{\frac{N_1}{N_2}}\ d(F_X(x)^{N_2}) \\
&+ (1-(\frac{1}{1+\delta})^{n_2}) - \frac{N_2}{N_1+N_2}(1-(\frac{1}{1+\delta})^{\frac{n_2(N_1+N_2)}{N_2}}) \\
&= \frac{N_1}{N_1+N_2} - \frac{n_1}{n_1+n_2}(\frac{1}{(1+\delta)})^{n_2} - \frac{n_2}{n_1+n_2}(1+\delta)^{n_1} d^{\frac{(n_1+n_2)N_2}{n_2}} + \frac{N_2}{N_1+N_2} d^{(N_1+N_2)}
\end{aligned}
$$

□

Here we see that the probability of paradox is 1 for all $\delta \geq [2(n_1/n_2+1)]^{1/n_1} - 1$. When $n_1 \gg n_2$, then $[2(n_1/n_2+1)]^{1/n_1} - 1 \approx n_1^{n_1} - 1$ and we see that the probability of paradox is 1 for extremely small values of δ.

Theorem 4.4. *Assume $g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1}(t) - g_{N_1,k} \circ g_{N_2,j}^{-1}(t)$ is non-negative when $t \in (0,1)$. Then*

$$
\begin{aligned}
\mathrm{P}\left(X_{1,(N_1-k)} > X_{2,(N_2-j)}, Y_{1,(n_1-k)} < Y_{2,(n_2-j)}\right) \leq (1+\delta)^p &\sum_{p=n_1-k}^{n_1} \left(\begin{array}{c} n_1 \\ p \end{array}\right) \frac{B(n_2-j+p, n_1-p+j+1)}{B(n_2-j, j+1)} \\
&- \sum_{p=N_1-k}^{N_1} \left(\begin{array}{c} N_1 \\ p \end{array}\right) \frac{B(N_2-j+p, N_1-p+j+1)}{B(N_2-j, j+1)}
\end{aligned}
$$

Proof.

$$
\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1-k)} > X_{2,(N_2-j)}, Y_{1,(n_1-k)} < Y_{2,(n_2-j)}\right) = \mathrm{P}\left(X_{2,(N_2-j)} < X_{1,(N_1-k)} < h_{1,2}\left(X_{2,(N_2-j)}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left[\int_x^{F_X^{-1} \circ g_{N_1,k}^{-1} \circ g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x)} f_{X_{1,(N_1-k)}}(y)\ dy\right]\ f_{X_{2,(N_2-j)}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left[\int_x^{F_X^{-1} \circ g_{N_1,k}^{-1} \circ g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x)} d\ g_{N_1,k}(F_X(y))\right]\ f_{X_{2,(N_2-j)}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left[g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x) - g_{N_1,k} \circ F_X(x)\right]^+\ f_{X_{2,(N_2-j)}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \left[g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1} \circ g_{N_2,j} \circ F_X(x) - g_{N_1,k} \circ F_X(x)\right]^+\ d\ g_{N_2,j}(F_X(x))
\end{aligned}
$$

$$
\begin{aligned}
&= \int_0^1 [g_{n_1,k} \circ F_{Y_1} \circ F_{Y_2}^{-1} \circ g_{n_2,j}^{-1}(t) - g_{N_1,k} \circ g_{N_2,j}^{-1}(t)]^+ \, dt \\
&= \int_0^1 g_{n_1,k}((1+\delta)x) \, dg_{n_2,j}(x) - \int_0^1 g_{N_1,k}(x) \, dg_{N_2,j}(x) \\
&= \int_0^1 I_{(1+\delta)x}(n_1-k,k+1) \frac{x^{n_2-j-1}(1-x)^j}{B(n_2-j,j+1)} \, dx - \int_0^1 I_x(N_1-k,k+1) \frac{x^{N_2-j-1}(1-x)^j}{B(N_2-j,j+1)} \, dx \\
&= \int_0^1 \sum_{p=n_1-k}^{n_1} \binom{n_1}{p} ((1+\delta)x)^p (1-(1+\delta)x)^{n_1-p} \frac{x^{n_2-j-1}(1-x)^j}{B(n_2-j,j+1)} \, dx \\
&- \int_0^1 \sum_{p=N_1-k}^{N_1} \binom{N_1}{p} x^p (1-x)^{N_1-p} \frac{x^{N_2-j-1}(1-x)^j}{B(N_2-j,j+1)} \, dx \\
&\leq (1+\delta)^p \int_0^1 \sum_{p=n_1-k}^{n_1} \binom{n_1}{p} x^p (1-x)^{n_1-p} \frac{x^{n_2-j-1}(1-x)^j}{B(n_2-j,j+1)} \, dx \\
&- \int_0^1 \sum_{p=N_1-k}^{N_1} \binom{N_1}{p} x^p (1-x)^{N_1-p} \frac{x^{N_2-j-1}(1-x)^j}{B(N_2-j,j+1)} \, dx \\
&= (1+\delta)^p \sum_{p=n_1-k}^{n_1} \binom{n_1}{p} \frac{B(n_2-j+p,n_1-p+j+1)}{B(n_2-j,j+1)} - \sum_{p=N_1-k}^{N_1} \binom{N_1}{p} \frac{B(N_2-j+p,N_1-p+j+1)}{B(N_2-j,j+1)}
\end{aligned}
$$

Example: Normal distributions

In this example, we demonstrate a case where Prop 4.2 applies. Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1$, and we will consider the following Full House Model for two systems,

$$
\begin{aligned}
Y_{1,j} &\sim N(\mu_1, \sigma_1^2), \qquad j = 1, \ldots, n_1, \\
Y_{2,j} &\sim N(\mu_2, \sigma_2^2), \qquad j = 1, \ldots, n_2, \\
X_{i,j} &\sim F_X(x), \qquad j = 1, \ldots, N_i,
\end{aligned}
$$

where $\mu_1 \leq \mu_2, \sigma_1 > 0$ and $\sigma_2 > 0$.

Theorem 4.5. *Assume* $\Phi(\frac{\Phi^{-1}(t^{\frac{N_2}{n_2}})\sigma_2 + \mu_2 - \mu_1}{\sigma_1})^{\frac{n_1}{N_1}} > t$ *for all* $t \in (0,1)$,

$$
\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma_1}, \ldots, \frac{\mu_2-\mu_1}{\sigma_1}, 0, \ldots, 0)) - \frac{N_2}{N_1+N_2}.
\end{aligned}
$$

The first n_1 *elements of* $\mathbf{w}$ *is* $\frac{\mu_2-\mu_1}{\sigma_1}$ *and the last* $n_2 - 1$ *elements are 0.*
$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ *with* (n_1+n_2-1)*-dimensional mean vector*

$$
\boldsymbol{\mu} = (0, \ldots, 0)
$$

and $(n_1+n_2-1)\times(n_1+n_2-1)$ *covariance matrix*

$$\mathbf{\Sigma}=\begin{bmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{bmatrix}$$

$$\Sigma_{11}=\begin{bmatrix}\frac{\sigma_2^2}{\sigma_1^2}+1 & \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2}\\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2}\\ \frac{\sigma_2^2}{\sigma_1^2} & \dots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2}+1\end{bmatrix}_{n_1\times n_1}, \Sigma_{12}^T=\Sigma_{21}=\begin{bmatrix}\frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\\ \vdots & \ddots & & \vdots\\ \vdots & & \ddots & \vdots\\ \frac{\sigma_2}{\sigma_1} & \dots & \dots & \frac{\sigma_2}{\sigma_1}\end{bmatrix}_{(n_2-1)\times n_1}$$

$$\Sigma_{22}=\begin{bmatrix}2 & 1 & \dots & 1\\ 1 & \ddots & \ddots & \vdots\\ \vdots & \ddots & \ddots & 1\\ 1 & \dots & 1 & 2\end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

Proof.

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)}>X_{2,(N_2)},Y_{1,(n_1)}<Y_{2,(n_2)}\right)=\mathrm{P}\left(X_{2,(N_2)}<X_{1,(N_1)}<h_{1,2}\left(X_{2,(N_2)}\right)\right)\\
&=\int_{-\infty}^{+\infty}\int_{x}^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\ dy\ f_{X_{2,(N_2)}}(x)\ dx\\
&=\int_{-\infty}^{+\infty}\int_{x}^{h_{1,2}(x)} N_1(F_X(y))^{N_1-1}\ dy\ N_2F_X(x)^{N_2-1}\ dx\\
&=\int_{-\infty}^{+\infty}\int_{x}^{F_X^{-1}(\Phi(\frac{\Phi^{-1}(F_X^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1})^{\frac{n_1}{N_1}})(x)} N_1F_X(y)^{N_1-1}\ dy\ N_2F_X(x)^{N_2-1}\ dx\\
&=\int_{-\infty}^{+\infty}((\Phi(\frac{(\Phi^{-1}(F_X(x)^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1}-F_X(x)^{N_1})\ N_2F_X(x)^{N_2-1}\ dx\\
&=\int_0^1(\Phi(\frac{\Phi^{-1}(t^{\frac{1}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1}-t^{\frac{N_1}{N_2}})\ dt\\
&=\int_0^1(\Phi(\frac{\Phi^{-1}(t^{\frac{1}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1}\ dt-\frac{N_2}{N_1+N_2}\\
&=\int_{-\infty}^{+\infty}\Phi(\frac{\sigma_2x+\mu_2-\mu_1}{\sigma_1})^{n_1}(n_2\Phi(x)^{n_2-1}\phi(x))\ dx-\frac{N_2}{N_1+N_2}\\
&=n_2\int_{-\infty}^{+\infty}\mathbb{P}(Z_{1,1}-\frac{\sigma_2}{\sigma_1}X\le\frac{\mu_2-\mu_1}{\sigma_1},\dots,Z_{1,n_1}-\frac{\sigma_2}{\sigma_1}X\le-\frac{\mu_2-\mu_1}{\sigma_1},\\
&Z_{2,1}-X\le0,\dots,Z_{2,n_2-1}-X\le0|X=x)\phi(x)\ dx-\frac{N_2}{N_1+N_2}\\
&=n_2\mathbb{P}(Z_{1,1}-\frac{\sigma_2}{\sigma_1}X\le\frac{\mu_2-\mu_1}{\sigma_1},\dots,Z_{1,n_1}-\frac{\sigma_2}{\sigma_1}X\le-\frac{\mu_2-\mu_1}{\sigma_1},Z_{2,1}-X\le0,\dots,Z_{2,n_2-1}-X\le0)
\end{aligned}$$

$$- \frac{N_2}{N_1 + N_2}$$
$$= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, \frac{\mu_2 - \mu_1}{\sigma_1}, 0, \ldots, 0)) - \frac{N_2}{N_1 + N_2}$$

where

$$\begin{aligned} h_{1,2}(x) &= (g_1 \circ f_2)(x) \\ &= F_X^{-1}((F_{Y_2}(F_{Y_1}^{-1}(F_X^{\frac{N_2}{n_2}})))^{\frac{n_1}{N_1}})(x) \\ &= F_X^{-1}((\Phi(\frac{\Phi^{-1}(F_X^{\frac{N_2}{n_2}})\sigma_2 + \mu_2 - \mu_1}{\sigma_1}))^{\frac{n_1}{N_1}})(x) \end{aligned}$$

The first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2 - \mu_1}{\sigma_1}$ and the last $n_2 - 1$ elements are 0.
$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $(n_1 + n_2 - 1)$-dimensional mean vector

$$\boldsymbol{\mu} = (0, \ldots, 0)$$

and $(n_1 + n_2 - 1) \times (n_1 + n_2 - 1)$ covariance matrix

$$\boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

$$\Sigma_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} + 1 & \frac{\sigma_2^2}{\sigma_1^2} & \ldots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ldots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2} + 1 \end{bmatrix}_{n_1 \times n_1}, \Sigma_{12}^T = \Sigma_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(n_2-1) \times n_1}$$

$$\Sigma_{22} = \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}_{(n_2-1) \times (n_2-1)}$$

□

Lemma 4.6. *Supposed* $\mathbf{X} \sim \mathcal{N}_n(\boldsymbol{\mu} = \mathbf{0}, \boldsymbol{\Sigma})$, $\boldsymbol{\mu}$ *and* $\boldsymbol{\Sigma}$ *are partitioned as follows,*

$$\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu_1} \\ \boldsymbol{\mu_2} \end{bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{0} \end{bmatrix} \text{ with sizes } \begin{bmatrix} 1 \times 1 \\ (n-1) \times 1 \end{bmatrix}$$
$$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix} \text{ with sizes } \begin{bmatrix} 1 \times 1 & 1 \times (n-1) \\ (n-1) \times 1 & (n-1) \times (n-1) \end{bmatrix}$$

Denote $\mathbf{y} = (y_1, y_2, \ldots, y_n)$ *and* $\mathbf{y}_{-1} = (y_2, \ldots, y_n)$
then

$$\frac{\partial}{\partial y_1}\mathbf{\Phi}_{\mathbf{X_n}}(\mathbf{y}, \mathbf{\Sigma}) = \phi(y_1) \cdot \mathbf{\Phi}_{\mathbf{X_{n-1}}}(\tilde{\mathbf{y}}, \tilde{\mathbf{\Sigma}})$$

where

$$\tilde{\mathbf{y}} = \mathbf{y}_{-1} - \mathbf{\Sigma}_{21}\mathbf{\Sigma}_{11}^{-1} \cdot y_1$$

and

$$\tilde{\mathbf{\Sigma}} = \mathbf{\Sigma}_{22} - \mathbf{\Sigma}_{21}\mathbf{\Sigma}_{11}^{-1}\mathbf{\Sigma}_{12}$$

Also,

$$\frac{\partial^2}{\partial y_i \partial y_j}\mathbf{\Phi}_{\mathbf{X_n}}(\mathbf{y}, \mathbf{\Sigma}) = \phi(y_i)\phi(y_j) \cdot \mathbf{\Phi}_{\mathbf{X_{n-2}}}(y_1, \ldots, y_{i-1}, y_{i+1}, \ldots, y_{j-1}, y_{j+1}, \ldots, y_n | y_i, y_j)$$

where $i \neq j$

Proof.

$$\frac{\partial}{\partial y_1}\mathbf{\Phi}_{\mathbf{X_n}}(\mathbf{y}, \mathbf{\Sigma}) = \phi(y_1) \cdot \mathbf{\Phi}_{\mathbf{X_{n-1}}}(X_2 = y_2, \ldots, X_n = y_n | X_1 = y_1)$$

Then

$$\frac{\partial^2}{\partial y_i \partial y_j}\mathbf{\Phi}_{\mathbf{X_n}}(\mathbf{y}, \mathbf{\Sigma}) = \phi(y_i)\phi(y_j) \cdot \mathbf{\Phi}_{\mathbf{X_{n-2}}}(y_1, \ldots, y_{i-1}, y_{i+1}, \ldots, y_{j-1}, y_{j+1}, \ldots, y_n | y_i, y_j)$$

where $i \neq j$

$\mathbf{\Phi}_{\mathbf{X_{n-1}}}(X_2 = y_2, \ldots, X_n = y_n | X_1 = y_1)$ is also a multivariate normal distribution with adjusted mean and covariance. Then using the Schur complement,

$$\tilde{\mu} = \boldsymbol{\mu}_2 + \mathbf{\Sigma}_{21}\mathbf{\Sigma}_{11}^{-1}(y_1 - \boldsymbol{\mu}_1) = \mathbf{\Sigma}_{21}\mathbf{\Sigma}_{11}^{-1} \cdot y_1$$

$$\tilde{\mathbf{y}} = \mathbf{y}_{-1} - \tilde{\mu} = \mathbf{y}_{-1} - \mathbf{\Sigma}_{21}\mathbf{\Sigma}_{11}^{-1} \cdot y_1$$

and

$$\tilde{\mathbf{\Sigma}} = \mathbf{\Sigma}_{22} - \mathbf{\Sigma}_{21}\mathbf{\Sigma}_{11}^{-1}\mathbf{\Sigma}_{12}$$

□

Supposed $\hat{\mu}_1, \hat{\mu}_2$ and $\hat{\sigma}_1$ are root sample-size consistent estimator, which indicates

$$|\hat{\mu}_1 - \mu_1| = Op(n_1^{-1/2}),$$

$$|\hat{\mu}_2 - \mu_2| = Op(n_2^{-1/2}),$$

$$|\hat{\sigma}_1 - \sigma_1| = Op(n_1^{-1/2}),$$

Notice that

$$\begin{aligned}
&\mathbf{\Phi_{W_{n_1+n_2-3}}}(W_2 = \theta, \ldots, W_{n_1} = \theta, W_{n_1+1} = 0, \ldots, W_{n_1+n_2-1} = 0 | W_1 = \theta) \\
&\quad = \mathbf{\Phi}_{W_{n_1+n_2-3}}(W_{n_1+n_2-1} = 0 | W_1 = \theta, W_2 = \theta, \ldots, W_{n_1} = \theta, W_{n_1+1} = 0, \ldots, W_{n_1+n_2-3} = 0) \times \ldots \\
&\times \mathbf{\Phi}_{W_1}(W_2 = \theta | W_1 = \theta)
\end{aligned}$$

$$\begin{aligned}
w_{max} = \max \Big\{ &\mathbf{\Phi_{W_{n_1+n_2-3}}}(W_{n_1+n_2-1} = 0 | W_1 = \theta, W_2 = \theta, \ldots, W_{n_1} = \theta, W_{n_1+1} = 0, \ldots, \\
&W_{n_1+n_2-3} = 0), \ldots, \mathbf{\Phi_{W_1}}(W_2 = \theta | W_1 = \theta) \Big\}
\end{aligned}$$

where $w_{max} \in (0, 1)$

$$\Phi(\mathbf{w}, \Sigma) = F_{\mathbf{W}}(\mathbf{W} = \mathbf{w}), \mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\mathbf{0}, \mathbf{\Sigma})$$

$$\mathbf{w} = (\frac{\mu_2 - \mu_1}{\sigma_1}, \ldots, \frac{\mu_2 - \mu_1}{\sigma_1}, 0, \ldots, 0)$$

$$\mathbf{\Sigma} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

$$\Sigma_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} + 1 & \frac{\sigma_2^2}{\sigma_1^2} & \ldots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ldots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2} + 1 \end{bmatrix}_{n_1 \times n_1}, \Sigma_{12}^T = \Sigma_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}_{(n_2-1) \times n_1}$$

$$\Sigma_{22} = \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}_{(n_2-1) \times (n_2-1)}$$

$$\begin{aligned}
\Phi(\hat{\mathbf{w}}, \hat{\Sigma}) - \Phi(\mathbf{w}, \Sigma) \approx &\frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_1}(\hat{\mu_1} - \mu_1) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \mu_2}(\hat{\mu_2} - \mu_2) + \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_1}(\hat{\sigma_1} - \sigma_1) \\
&+ \frac{\partial \Phi(\mathbf{w}, \Sigma)}{\partial \sigma_2}(\hat{\sigma_2} - \sigma_2)
\end{aligned}$$

$$\begin{aligned}\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \mu_1} &= \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial w_i} \cdot \frac{\partial w_i}{\partial \mu_1} \\ &= -\frac{1}{\sigma_1} \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial w_i} \\ &= -\frac{n_1}{\sigma_1} \cdot \phi(w_1) \cdot \Phi(\tilde{\mathbf{w}},\tilde{\Sigma})\end{aligned}$$

$$\begin{aligned}\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \mu_2} &= \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial w_i} \cdot \frac{\partial w_i}{\partial \mu_2} \\ &= \frac{1}{\sigma_1} \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial w_i} \\ &= \frac{n_1}{\sigma_1} \cdot \phi(w_1) \cdot \Phi(\tilde{\mathbf{w}},\tilde{\Sigma})\end{aligned}$$

$$\begin{aligned}\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_1} &= \sum_{i=1}^{n_1} \frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial w_i} \cdot \frac{\partial w_i}{\partial \sigma_1} + tr(\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1}) \\ &= -\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1) \cdot \Phi(\tilde{\mathbf{w}},\tilde{\Sigma}) + tr(\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma} \cdot \frac{\partial \Sigma}{\partial \sigma_1})\end{aligned}$$

Define the $\phi(\mathbf{x})$ is the probability density function of $\Phi(\mathbf{w},\Sigma)$

$$\begin{aligned}\frac{\partial \log(\phi(\mathbf{x}))}{\partial \Sigma} &= -\frac{1}{2}(\frac{\partial \log(det(\Sigma))}{\partial \Sigma} + \frac{\partial \mathbf{x}^t\Sigma^{-1}\mathbf{x}}{\partial \Sigma}) \\ &= -\frac{1}{2}(\Sigma^{-1} - \Sigma^{-1}\mathbf{x}\mathbf{x}^t\Sigma^{-1})\end{aligned}$$

$$\begin{aligned}\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma} &= \frac{\partial \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x})d\mathbf{x}}{\partial \Sigma} \\ &= \int_{-\infty}^{\mathbf{w}} \frac{\partial \phi(\mathbf{x})}{\partial \Sigma} d\mathbf{x} \\ &= \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \frac{\partial \log(\phi(\mathbf{x}))}{\partial \Sigma} d\mathbf{x} \\ &= -\frac{1}{2}\Sigma^{-1}\Phi(\mathbf{w},\Sigma) + \frac{1}{2}\Sigma^{-1}\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} \\ &= \frac{1}{2}\Sigma^{-1}\left(\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x}\Sigma^{-1} - \Phi(\mathbf{w},\Sigma)\right)\end{aligned}$$

$$\frac{\partial \Sigma}{\partial \sigma_1} = \begin{bmatrix} \frac{\partial \Sigma_{11}}{\partial \sigma_1} & \frac{\partial \Sigma_{12}}{\partial \sigma_1} \\ \frac{\partial \Sigma_{21}}{\partial \sigma_1} & \frac{\partial \Sigma_{22}}{\partial \sigma_1} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

$$A_{11} = \left[-\frac{2\sigma_2^2}{\sigma_1^3}\right]_{n_1 \times n_1}, A_{12}^T = A_{21} = \left[-\frac{\sigma_2}{\sigma_1^2}\right]_{(n_2-1)\times n_1}, A_{22} = \left[0\right]_{(n_2-1)\times(n_2-1)}$$

By applying Schur complement,

$$\mathbf{\Sigma}^{-1} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}^{-1} = \begin{bmatrix} (\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} & -(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1} \\ -\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} & \Sigma_{22}^{-1} + \Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1} \end{bmatrix}$$

$$B_{11} = (\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} = \frac{1}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2} + 1 & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} \\ -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \cdots & -\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2} & \frac{\sigma_2^2}{\sigma_1^2}\frac{n_1-1}{n_2} + 1 \end{bmatrix}_{n_1 \times n_1}$$

$$B_{12} = -(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1} = \left[-\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\right]_{n_1 \times (n_2-1)}$$

$$B_{21} = -\Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1} = \left[-\frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1}\right]_{(n_2-1)\times n_1}$$

$$\begin{aligned} B_{22} =& \Sigma_{22}^{-1} + \Sigma_{22}^{-1}\Sigma_{21}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})^{-1}\Sigma_{12}\Sigma_{22}^{-1} \\ =& \begin{bmatrix} 1 - \frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \\ -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & \cdots & -\frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} & 1 - \frac{1}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)n_2} \end{bmatrix}_{(n_2-1)\times(n_2-1)} \end{aligned}$$

Based on the structure of the Σ^{-1} and $\frac{\partial \Sigma}{\partial \sigma_1}$, we have

$$tr(\Sigma^{-1}\frac{\partial \Sigma}{\partial \sigma_1})$$

$$
\begin{aligned}
&= tr(\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}) \\
&= tr(B_{11}A_{11} + B_{12}A_{21}) + tr(B_{21}A_{12} + B_{22}A_{22}) \\
&= n_1 \left(\frac{-\frac{2\sigma_2^2}{\sigma_1^3}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1} (\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1 - 1}{n_2} + 1 - (n_1 - 1)\frac{\sigma_2^2}{\sigma_1^2}\frac{1}{n_2}) + \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1}\frac{\sigma_2}{\sigma_1^2}(n_2 - 1) \right) \\
&+ (n_2 - 1) \left(n_1 \frac{\frac{\sigma_2}{\sigma_1}\frac{1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1}\frac{\sigma_2}{\sigma_1^2} \right) \\
&= -2\frac{\frac{\sigma_2^2}{\sigma_1^3}\frac{n_1}{n_2}}{\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2} + 1}
\end{aligned}
$$

$$
\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} = \begin{bmatrix} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1^2 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1 x_n d\mathbf{x} \\ \vdots & \ddots & \vdots \\ \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n x_1 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n^2 d\mathbf{x} \end{bmatrix}
$$

Define $\mathbf{w}_{-i}$ as the vector $\mathbf{w}$ with the ith component removed, and $\mathbf{w}_{-(i,j)}$ as the vector $\mathbf{w}$ with the ith and j th components removed. Similarly, $\mathbf{x}_{-i}$ and $\mathbf{x}_{-(i,j)}$ are defined in the same manner.

Then

$$
\begin{aligned}
\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_i^2 d\mathbf{x} &= \int_{-\infty}^{w_i} \left(\int_{-\infty}^{\mathbf{w}_{-i}} \phi(\mathbf{x}_{-i}|x_i) d\mathbf{x}_{-i} \right) \; x_i^2 \; \phi(x_i) dx_i \\
&\leq w_{max}^{n_1+n_2-3} \cdot Var(X_i) \\
&\leq w_{max}^{n_1+n_2-3} \cdot Var(X_i)
\end{aligned}
$$

$$
\begin{aligned}
\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_i x_j d\mathbf{x} &= \int_{-\infty}^{w_i} \int_{-\infty}^{w_j} \left(\int_{-\infty}^{\mathbf{w}_{-(i,j)}} \phi(\mathbf{x}_{-(i,j)}|x_i, x_j) d\mathbf{x}_{-(i,j)} \right) \; x_i x_j \; \phi(x_i, x_j) dx_i dx_j \\
&\leq w_{max}^{n_1+n_2-3} \cdot Cov(X_i, X_j)
\end{aligned}
$$

$$
\begin{aligned}
\int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot \mathbf{x}\mathbf{x}^t d\mathbf{x} &= \begin{bmatrix} \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1^2 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_1 x_n d\mathbf{x} \\ \vdots & \ddots & \vdots \\ \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n x_1 d\mathbf{x} & \dots & \int_{-\infty}^{\mathbf{w}} \phi(\mathbf{x}) \cdot x_n^2 d\mathbf{x} \end{bmatrix} \\
&\leq \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}
\end{aligned}
$$

$$S_{11} = \begin{bmatrix} w_{max}^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}+1) & w_{max}^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \dots & w_{max}^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) \\ w_{max}^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & w_{max}^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) \\ w_{max}^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & \dots & w_{max}^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}) & w_{max}^{n_1+n_2-3}(\frac{\sigma_2^2}{\sigma_1^2}+1) \end{bmatrix}_{n_1\times n_1}$$

$$S_{12}^T = S_{21} = \begin{bmatrix} w_{max}^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) & \dots & \dots & w_{max}^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ w_{max}^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) & \dots & \dots & w_{max}^{n_1+n_2-3}(\frac{\sigma_2}{\sigma_1}) \end{bmatrix}_{(n_2-1)\times n_1}$$

$$S_{22} = \begin{bmatrix} 2w_{max}^{n_1+n_2-3} & w_{max}^{n_1+n_2-3} & \dots & w_{max}^{n_1+n_2-3} \\ w_{max}^{n_1+n_2-3} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & w_{max}^{n_1+n_2-3} \\ w_{max}^{n_1+n_2-3} & \dots & w_{max}^{n_1+n_2-3} & 2w_{max}^{n_1+n_2-3} \end{bmatrix}_{(n_2-1)\times(n_2-1)}$$

$$tr(\Sigma^{-1}\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\Sigma^{-1}\frac{\partial\Sigma}{\partial\sigma_1})$$

$$= tr(\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix})$$

$$= tr(\begin{bmatrix} B_{11}S_{11}+B_{12}S_{21} & B_{11}S_{12}+B_{12}S_{22} \\ B_{21}S_{11}+B_{22}S_{21} & B_{21}S_{12}+B_{22}S_{22} \end{bmatrix}\begin{bmatrix} B_{11}A_{11}+B_{12}A_{21} & B_{11}A_{12}+B_{12}A_{22} \\ B_{21}A_{11}+B_{22}A_{21} & B_{21}A_{12}+B_{22}A_{22} \end{bmatrix})$$

$$= tr((B_{11}S_{11}+B_{12}S_{21})(B_{11}A_{11}+B_{12}A_{21})+(B_{11}S_{12}+B_{12}S_{22})(B_{21}A_{11}+B_{22}A_{21}))$$

$$+ tr((B_{21}S_{11}+B_{22}S_{21})(B_{11}A_{12}+B_{12}A_{22})+(B_{21}S_{12}+B_{22}S_{22})(B_{21}A_{12}+B_{22}A_{22}))$$

$$= -\frac{\frac{n_1\sigma_2^2}{\sigma_2^2}(\frac{2n_1\sigma_2^2}{\sigma_2^3}+1)}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}w_{max}^{n_1+n_2-3} + \frac{\frac{n_1\sigma_2^2}{\sigma_2^2}\frac{n_2-1}{n_2}(\frac{n_1\sigma_2^2}{\sigma_2^3}+1)}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}w_{max}^{n_1+n_2-3} + \frac{\frac{2n_1^2\sigma_2^4}{\sigma_2^5}\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}w_{max}^{n_1+n_2-3}$$

$$-\frac{\frac{n_1^2\sigma_2^4}{\sigma_2^5}(\frac{n_2-1}{n_2})^2}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}w_{max}^{n_1+n_2-3} + \frac{\frac{2n_1^2\sigma_2^4}{\sigma_2^5}\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}w_{max}^{n_1+n_2-3} + \frac{\frac{n_1(n_2-1)\sigma_2^2}{\sigma_2^3}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}-1)w_{max}^{n_1+n_2-3}$$

$$-\frac{\frac{2n_1^2\sigma_2^4}{\sigma_2^5}\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}w_{max}^{n_1+n_2-3} + \frac{\frac{n_1(n_2-1)\sigma_2^2}{\sigma_2^3}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(1-\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)})w_{max}^{n_1+n_2-3} + \frac{\frac{n_1\sigma_2^2}{\sigma_2^3}\frac{n_2-1}{n_2}(n_1\frac{\sigma_2^2}{\sigma_1^2}+1)}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}w_{max}^{n_1+n_2-3}$$

$$+\frac{\frac{n_1(n_2-1)\sigma_2^2}{\sigma_2^3}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}-1)w_{max}^{n_1+n_2-3} - \frac{\frac{n_1^2\sigma_2^4}{\sigma_2^5}(\frac{n_2-1}{n_2})^2}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)^2}w_{max}^{n_1+n_2-3} + \frac{\frac{n_1(n_2-1)\sigma_2^2}{\sigma_2^3}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)}(1-\frac{\frac{n_2-1}{n_2}}{(\frac{\sigma_2^2}{\sigma_1^2}\frac{n_1}{n_2}+1)})w_{max}^{n_1+n_2-3}$$

$$= OP(n_1^2 w_{max}^{n_1+n_2-3} + n_1 n_2 w_{max}^{n_1+n_2-3})$$

Then

$$\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_1} = -\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma}) + tr(\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma}\cdot\frac{\partial \Sigma}{\partial \sigma_1})$$
$$= -\frac{(\mu_2-\mu_1)n_1}{\sigma_1^2}\phi(w_1)\cdot\Phi(\tilde{\mathbf{w}},\tilde{\Sigma}) + OP(n_1^2 w_{max}^{n_1+n_2-3} + n_1 n_2 w_{max}^{n_1+n_2-3})$$

Similarly,

$$\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_2} = tr(\frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \Sigma}\cdot\frac{\partial \Sigma}{\partial \sigma_2})$$
$$= OP(n_1^2 w_{max}^{n_1+n_2-3} + n_1 n_2 w_{max}^{n_1+n_2-3})$$

$$\Phi(\hat{\mathbf{w}},\hat{\Sigma}) - \Phi(\mathbf{w},\Sigma) \approx \frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \mu_1}(\hat{\mu_1}-\mu_1) + \frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \mu_2}(\hat{\mu_2}-\mu_2) + \frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_1}(\hat{\sigma_1}-\sigma_1)$$
$$+ \frac{\partial \Phi(\mathbf{w},\Sigma)}{\partial \sigma_2}(\hat{\sigma_2}-\sigma_2)$$
$$= OP(n_1^{\frac{1}{2}} w_{max}^{n_1+n_2-2} + n_1 n_2^{-\frac{1}{2}} w_{max}^{n_1+n_2-2} + n_1^{\frac{3}{2}} w_{max}^{n_1+n_2-3} + n_1^{\frac{1}{2}} n_2 w_{max}^{n_1+n_2-3}$$
$$+ n_1^2 n_2^{-\frac{1}{2}} w_{max}^{n_1+n_2-3} + n_1 n_2^{\frac{1}{2}} w_{max}^{n_1+n_2-3})$$

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \mu_1.\mu_2, \sigma_1, \sigma_2\right) - \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)} | \hat{\mu_1}.\hat{\mu_2}, \hat{\sigma_1}, \hat{\sigma_2}\right)$$
$$= OP(n_1^{\frac{1}{2}} n_2 w_{max}^{n_1+n_2-2} + n_1 n_2^{\frac{1}{2}} w_{max}^{n_1+n_2-2} + n_1^{\frac{3}{2}} n_2 w_{max}^{n_1+n_2-3} + n_1^{\frac{1}{2}} n_2^2 w_{max}^{n_1+n_2-3} + n_1^2 n_2^{\frac{1}{2}} w_{max}^{n_1+n_2-3} + n_1 n_2^{\frac{3}{2}} w_{max}^{n_1+n_2-3})$$

5 Examples

5.1 Theory supporting simulations

In this section we verify Theorem 3.4, Theorem 4.3, and 4.5 in the numerical simulation. Directly paradox probabilities from these theorems are compared with Monte Carlo paradox probabilities in a variety of settings. Each of these settings will involve comparisons of the observed statistics and latent aptitudes of the highest achievers, the pair $(Y_{i,(n_i)}, X_{i,(N_i)})$ across two populations.

The simulations will proceed as follows: Distributions for Y_1 and Y_2 are specified along with sample sizes n_i, N_i, $i = 1, 2$. We will specify a Monte Carlo sample size of 100. For iteration m of the Monte Carlo sample procedure $Y_{1,j}$, $j = 1, \ldots, n_1$ and $Y_{2,k}$, $k = 1, \ldots, n_2$ are generated. Then $X_{i,(N_i)}$, $i = 1, 2$ is calculated using (1) where $X \sim \text{Pareto}(\alpha = 1.16)$. Then $1(X_{(N_1)} > X_{(N_2)}, Y_{(n_1)} < Y_{(n_2)})$ is recorded.

The results are tabulated in Theorem 1. These results show strong agreement between our theory and numerical simulation. They also reveal that paradox probabilities are non-trivial across these configurations.

Setting	Distribution for Y_1	Distribution for Y_2	Prob from Theorem	Prob from simulation
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 40$	$N(0,1), n_2 = 60$	0.1495	0.1472
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 30$	$N(0,1), n_2 = 70$	0.2866	0.2875
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 60$	$N(0,1), n_2 = 40$	0	0
$N_1 = 1e3, N_2 = 1e3, k = n_1 - 1, j = n_2 - 1$	$N(0,1), n_1 = 70$	$N(0,1), n_2 = 30$	0	0
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 40$	$U(0,1+\delta), n_2 = 60$	0.1233	0.1224
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 30$	$U(0,1+\delta), n_2 = 70$	0.2203	0.2210
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 60$	$U(0,1+\delta), n_2 = 40$	0.0022	0.0021
$N_1 = 1e5, N_2 = 1e5, k = n_1, j = n_2$	$U(0,1), n_1 = 70$	$U(0,1+\delta), n_2 = 30$	0.0008	0.0007
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 40$	$N(1,4), n_2 = 60$	0.3338	0.3356
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 30$	$N(1,4), n_2 = 70$	0.3336	0.3343
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 10$	$N(1,4), n_2 = 100$	0.3331	0.3345
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,2), n_1 = 100$	$N(1,4), n_2 = 10$	0.2378	0.2406
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 40$	$N(1,1), n_2 = 60$	0.3321	0.3326
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 30$	$N(1,1), n_2 = 70$	0.3313	0.3325
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 10$	$N(1,1), n_2 = 100$	0.3334	0.3346
$N_1 = 1e5, N_2 = 2e5, k = n_1, j = n_2$	$N(-1,1), n_1 = 100$	$N(1,1), n_2 = 10$	0.2593	0.2592

Table 1: Simulation results to validate the theory

5.2 Recursive Full House Models

In this section we will investigate a simple recursive Full House Models in which Full House Models are repeatedly applied to individuals as they progress through iterations of some generic process. The general setup is that, in the first round, latent aptitude $X_j \overset{iid}{\sim} F_X$, $j = 1, \ldots, N$ will be generated and then a label will be randomly assigned to each individual. There will be m labels and each label will designate the outcome distribution. $F_{Y_{i,j}}$, $i = 1, \ldots, n$, $j = 1, \ldots, n_i$. Then outcomes will be realized as

$$Y_{i,(r_{i,(j)})} = F^{-1}_{Y_{i,j}} \circ I^{-1}_{r_{i,(j)}, n_i - r_{i,(j)}+1} \circ I_{j,N-j+1} \circ F_X(X_{(j)}), \tag{6}$$

where $r_{i,(j)}$ is the rank of $X_{(j)}$ among the n_i latent aptitudes also assigned label i.

In the next iteration, all observations will be combined and then sorted in descending order by $Y_{i,(j)}$. The labels will be assigned in descending order, the largest n_m values of $Y_{i,(j)}$ are assigned label m, the next largest n_{m-1} values of $Y_{i,(j)}$ are assigned label $m-1$, and so on. We will setup these simulation so that Y_i stochastically dominates Y_j whenever $i > j$ so that higher outcomes are associated with higher labels. Then new Y values are generated according to (6). The process will repeat until a pre-assigned stopping rule is satisfied. Note that this simulation setup assumes an immutable latent aptitude.

5.2.1 Stop when highest latent aptitude is included in the highest-performing group

Here we consider a recursive Full House Modeling setup described in Section 5.2 in which the recursion stops at the first iteration when $X_{(N)}$ is assigned label m. Here we specify that $N = 100,000$, $m = 200$, $n_i = 500$, $i = 1, \ldots, m$. We will assume that $X_j \overset{iid}{\sim}$ Pareto(1.16), $j = 1, \ldots, N$. And we will assume that $Y_{i,j} \overset{iid}{\sim} N(\mu_i, \sigma_i)$, $j = 1, \ldots, n_i$ and $i = 1, \ldots, m$. We will specify that the μ_is as a set of 200 equally spaced values between 60 and 80 with μ_i increasing in i, ie $\mu_i = 60 - 20/199 + 20i/199$, $i = 1, \ldots, m$. We will fix $\sigma_i = \sigma$ and consider $\sigma \in \{0.5, 1, 2\}$. These simulation will be repeated 500 times.

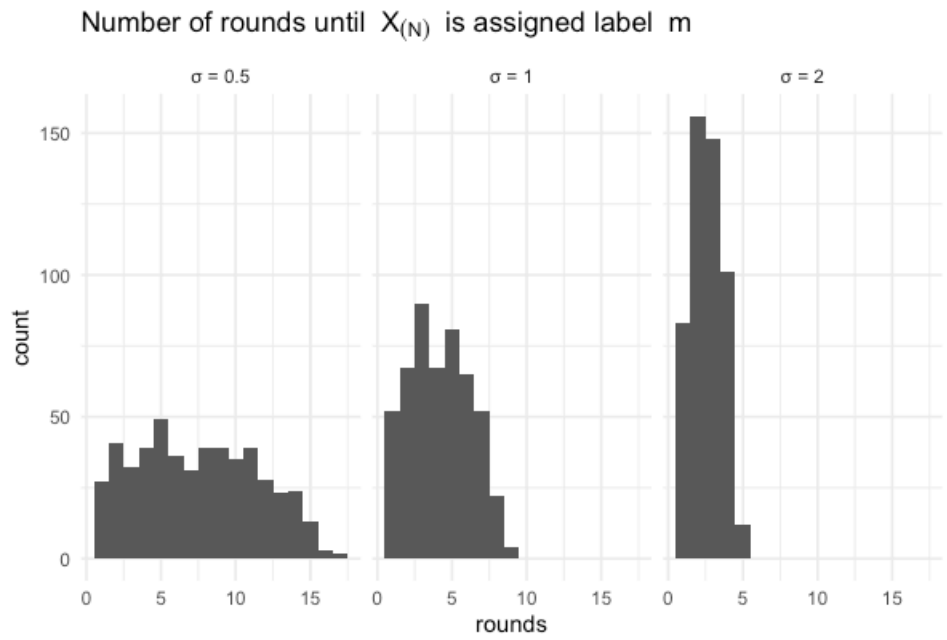


Figure 1: The number of rounds that it takes for $X_{(N)}$ to be assigned label m in a recursive Full House Modeling framework in Section 5.2 with specifications in Section 5.2.1.

To illustrate the procedure, consider a simplified scenario in which all $X_{(j)}$ values are realized at their median value. With this simplification, $Y_{i,(r_{i,(j)})}$ in (6) is approximately $F^{-1}_{Y_{i,j}}((r_{i,(j)} - 1/3)/(n_i + 1/3))$ [Kerman, 2011]. And we have that $Y_{i,(n_i)} - Y_{i+1,(n_{i+1}-1)} > \mu_{i+1} - \mu_i$ for all considered values of σ. The individual with latent ability $X_{(N)}$ will consistently attain the value $Y_{i,(n_i)}$, regardless their assigned group. Consequently, this individual's position among the m groups will improve as the simulation advances.

Results can be seen in Figure 1. This Monte Carlo simulation reveal that it is expected to take several rounds before the individual with the top aptitude belongs to the group that allows for the highest performances. On average, with σ set to 0.5, 1, and 2, it takes 2.61, 4.19, and 7.46 rounds, respectively, for the individual with the top aptitude to join the highest-performing group (respective standard errors: 0.05, 0.09, 0.18).

Notes:

Waters and Miikkulainen [2014] developed an automated machine learning based admissions systems, called GRADE, designed to save time by pre-classifying people who are likely to be admitted to the Department of Computer Science at the University of Texas at Austin. This procedure ignored statements of purpose because, in practice, it was found that features in a statement of purpose were not discriminative and did not improve the quality of GRADE's predictions.

Alvero et al. [2020] used a "unique corpus of 283,676 application essays submitted to a large, selective, state university system between 2015 and 2016, we assess the extent to which applicant demographic characteristics can be inferred from application essays." They found that the word *baseball* best classified Male applicants.

Baker and Hawn [2022] reviewed algorithmic bias in education. They concluded that "there are a great number of "known unknowns" in the study of algorithmic bias in education – areas where we know that our current knowledge is insufficient – but also exposes the possibility that there may also be many "unknown unknowns" – groups that we do not even realize are being severely impacted. Foremost among these "unknown unknowns" is the lack of empirical work on how algorithmic bias impact specific groups of learners outside the United States, at a finer grain-size than the national level."

Both meta-analyses in Walton and Spencer [2009] revealed a bias in standard measures of academic performance–both test scores and classroom grades. The bias results from psychological threat. It causes measures of academic performance to underestimate the true intellectual ability and potential of ethnic minority students and of women in quantitative fields.

5.3 Batting averages in Major League Baseball

In this analysis, we compute the probability of the paradox using the results from the Normal distribution example above and illustrate it in the application of Major League Baseball (MLB) since it is widely acknowledged that the batting average follows Normal distribution.

Specifically, we compute the probability that the top batting average in 2023 is lower than the top batting average in the other season, but the players who have the highest batting average in 2023 are more talented than those who have the highest batting average in the other season.

In this analysis, we detail the considerations made for the batting average statistics that we era-adjust using the Full House Model [Eck, 2020b]. Park factor adjustment is considered in this analysis and we apply the adjusted park index from Schell [2016] to all ballparks from the 1871 season to the 2022 season. The batting average can be affected by the ballpark and we apply the park factor adjustment to this statistic, which accounts for the handedness of batters.

The Full House Model will only be applied to the statistics obtained by full-time players. We define the full-time hitter cutoff as the median PAs after screening out hitters who appeared in fewer than 75 PAs. This criterion is flexible enough to account for changing the number of games played over time as well as seasons shortened by labor strikes and pandemics.

In the 1994 and 1995[1], hitters and pitchers played fewer games per season than the regular season. To deal with extreme statistics from the small sample size, we employed a shrinkage method that adjusted the raw statistics toward a global average. Our shrinkage method follows a shrinkage of ballpark effect estimates motivated from Schell [2016]. The details can be found in Eck [2020b].

We compute the talent scores for the batting average in each season using the (**??**). Then the theoretical results from the Normal distribution example above can be applied to this example and compute the paradox probability of all the seasons from 1871 to 2022 compared with 2023. Fig 2 displays the tendency of the paradox probability of all the seasons from

[1]https://en.wikipedia.org/wiki/1994%E2%80%9395_Major_League_Baseball_strike

1871 to 2022 compared with 2023.

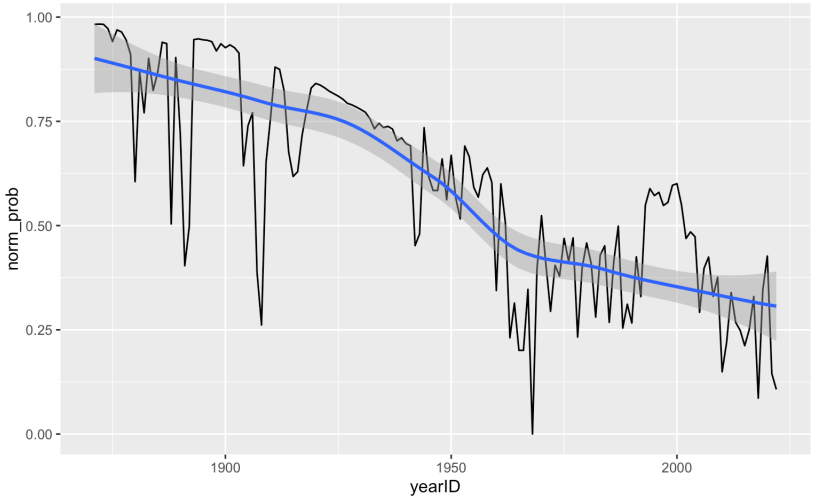


Figure 2: The paradox probability of all the seasons from 1871 to 2022 compared with 2023

Supplemental Results

Lemma 5.1. *Let $1 \leq j < n$ and let $x \in (0,1)$. Let $I_{\alpha,\beta}(x)$ be a regularized incomplete beta function as in Definition 3.3. Then, $I_{n-j,j+1}(x^{\frac{1}{n}}) \geq I_{n-j+1,j+1}(x^{\frac{1}{n+1}})$.*

Need to fix notation; should be $I_{\alpha,\beta}(x)$

Proof. We can define a function $f_{n,j}(x)$ as

$$\begin{aligned} f_{n,j}(x) &= I_{n-j,j+1}(x^{\frac{1}{n}}) - I_{n-j+1,j+1}(x^{\frac{1}{n+1}}) \\ &= \int_0^{x^{\frac{1}{n}}} \frac{y^{n-j-1}(1-y)^j}{B(n-j,j+1)}dy - \int_0^{x^{\frac{1}{n+1}}} \frac{y^{n-j}(1-y)^j}{B(n-j+1,j+1)}dy. \end{aligned}$$

An application of the chain rule yields

$$\begin{aligned} f'_{n,j}(x) &= \frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{nB(n-j,j+1)} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{(n+1)B(n-j+1,j+1)} \\ &= \frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{nB(n-j,j+1)} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{(n-j)B(n-j,j+1)} \\ &= \frac{1}{B(n-j,j+1)}\left(\frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^j\left(x^{\frac{1-n}{n}}\right)}{n} - \frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^j\left(x^{-\frac{n}{n+1}}\right)}{n-j}\right), \end{aligned}$$

where we used $B(n-j+1,j+1)=\frac{n-j}{n+1}B(n-j,j+1)$ in the second equality. Now define $A_{n,j}=\{x: f'_{n,j}(x)\geq 0\}$. For $x\in A_{n,j}$ we have

$$\begin{aligned}
&\frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^{j}\left(x^{\frac{1-n}{n}}\right)}{n}\geq\frac{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^{j}\left(x^{-\frac{n}{n+1}}\right)}{n-j}\\
\implies&\frac{\left(x^{\frac{1}{n}}\right)^{n-j-1}\left(1-x^{\frac{1}{n}}\right)^{j}\left(x^{\frac{1-n}{n}}\right)}{\left(x^{\frac{1}{n+1}}\right)^{n-j}\left(1-x^{\frac{1}{n+1}}\right)^{j}\left(x^{-\frac{n}{n+1}}\right)}\geq\frac{n}{n-j}\\
\implies&x^{-\frac{j}{n(n+1)}}\left(\frac{1-x^{\frac{1}{n}}}{1-x^{\frac{1}{n+1}}}\right)^{j}\geq\frac{n}{n-j}\\
\implies&\frac{x^{-\frac{1}{n(n+1)}}-x^{\frac{1}{n+1}}}{1-x^{\frac{1}{n+1}}}\geq\left(\frac{n}{n-j}\right)^{\frac{1}{j}}.
\end{aligned}$$

Define $g_n(x)=\frac{x^{-\frac{1}{n(n+1)}}-x^{\frac{1}{n+1}}}{1-x^{\frac{1}{n+1}}}$. Then

$$g_n(x)=\frac{x^{-\frac{1}{n(n+1)}}-x^{\frac{1}{n+1}}}{1-x^{\frac{1}{n+1}}}=\frac{x^{-\frac{1}{n(n+1)}}-1}{1-x^{\frac{1}{n+1}}}+1,$$

and

$$\begin{aligned}
g_n'(x)&=\frac{1}{(1-x^{\frac{1}{n+1}})^2}\left(-\frac{1}{n(n+1)}x^{-\frac{1}{n(n+1)}-1}(1-x^{\frac{1}{n+1}})-(x^{-\frac{1}{n(n+1)}}-1)(-\frac{1}{n+1}x^{\frac{1}{n+1}-1})\right)\\
&=\frac{1}{(1-x^{\frac{1}{n+1}})^2}\left(\frac{1}{n}x^{-\frac{n^2+1}{n(n+1)}}-\frac{1}{n+1}x^{-\frac{n}{n+1}}-\frac{1}{n(n+1)}x^{-\frac{n^2+n+1}{n(n+1)}}\right)\\
&=\frac{x^{-\frac{n^2+1}{n(n+1)}}}{n(n+1)(1-x^{\frac{1}{n+1}})^2}\left((n+1)-nx^{\frac{1}{n(n+1)}}-x^{-\frac{1}{n+1}}\right).
\end{aligned}$$

Now define $l_n(x)=(n+1)-nx^{\frac{1}{n(n+1)}}-x^{-\frac{1}{n+1}}$ and see that

$$l_n'(x)=-\frac{1}{n+1}x^{\frac{1}{n(n+1)}-1}+\frac{1}{n+1}x^{-\frac{1}{n+1}-1}=\frac{1}{n+1}x^{-\frac{n+2}{n+1}}\left(1-x^{\frac{1}{n}}\right).$$

Thus, $l_n(x)$ is an increasing function for $x\in(0,1)$ and $\sup_{x\in(0,1)}l_n(x)=0$. Therefore, $l_n(x)<0$ for $x\in(0,1)$. Thus, $g_n(x)$ is an decreasing function, and $g_n(x)\to 1$ as $x\to 1$. We can see that $g_n(x)\in(1,+\infty)$ for $x\in(0,1)$. Thus $A_{n,j}=(0,x^*_{n,j}]$ where $x^*_{n,j}$ is such that $g_n(x^*_{n,j})=(\frac{n}{n-j})^j$. Putting it all together, $f_{n,j}(0)=f_{n,j}(1)=0$ and this implies that $I_{n-j,j+1}(x^{\frac{1}{n}})\geq I_{n-j+1,j+1}(x^{\frac{1}{n+1}})$ for all $x\in(0,1)$. □

Proposition 5.2. *Consider a Full House Model in Definition 2.1 with two populations and $F_{Y_i}=F_Y$ for $i=1,2$ where the system inclusion function is $1_i(X_{i,j}\geq X_{i,(N_i-n_i+1)})$. Let I be a regularized incomplete beta function. In the notation of Theorem 3.4, suppose that $\frac{n_1}{N_1}\leq\frac{n_2}{N_2}$, $N_2-N_1>n_2-n_1\geq 0$, and $k=j$. Then, technical condition (4) holds and the conclusion of Theorem 3.4 follows.*

Proof. The conditions of this proposition allow us to express technical condition (4) as

$$f_{n_1,n_2,N_1,N_2,j}(x) = I_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x) - I_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x).$$

Applications of the chain rule, the inverse function theorem [Rudin et al., 1964] on Page 221, and this property of regularized incomplete beta functions

$$I'_{\alpha,\beta}(I^{-1}_{\alpha,\beta}(x)) = \frac{I^{-1}_{\alpha,\beta}(x)^{\alpha-1}(1-I^{-1}_{\alpha,\beta}(x))^{\beta-1}}{B(\alpha,\beta)},$$

allow us to write

$$\begin{aligned}
f'_{n_1,n_2,N_1,N_2,j}(x) &= \frac{I'_{n_1-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x)}{I'_{n_2-j,j+1} \circ I^{-1}_{n_2-j,j+1}(x)} - \frac{I'_{N_1-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x)}{I'_{N_2-j,j+1} \circ I^{-1}_{N_2-j,j+1}(x)} \\
&= \frac{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-j-1}\left(1-I^{-1}_{n_2-j,j+1}(x)\right)^{j}/B(n_1-j,j+1)}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_2-j-1}\left(1-I^{-1}_{n_2-j,j+1}(x)\right)^{j}/B(n_2-j,j+1)} \\
&\quad - \frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-j-1}\left(1-I^{-1}_{N_2-j,j+1}(x)\right)^{j}/B(N_1-j,j+1)}{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_2-j-1}\left(1-I^{-1}_{N_2-j,j+1}(x)\right)^{j}/B(N_2-j,j+1)} \\
&= \frac{B(n_2-j,j+1)}{B(n_1-j,j+1)}\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2} - \frac{B(N_2-j,j+1)}{B(N_1-j,j+1)}\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2} \\
&= \left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}\frac{B(N_2-j,j+1)}{B(N_1-j,j+1)} \\
&\quad \times \left[\frac{B(n_2-j,j+1)/B(n_1-j,j+1)}{B(N_2-j,j+1)/B(N_1-j,j+1)} - \frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}\right]
\end{aligned}$$

From here we see that $f_{n_1,n_2,N_1,N_2,j}(0) = f_{n_1,n_2,N_1,N_2,j}(1) = 0$, and when $x = 1$, we have

$$\begin{aligned}
f'_{n_1,n_2,N_1,N_2,j}(1) &= \frac{B(n_2-j,j+1)}{B(n_1-j,j+1)} - \frac{B(N_2-j,j+1)}{B(N_1-j,j+1)} \\
&= \frac{(n_1-j)\ldots(n_1)}{(n_2-j)\ldots(n_2)} - \frac{(N_1-j)\ldots(N_1)}{(N_2-j)\ldots(N_2)} \\
&\leq 0,
\end{aligned}$$

from the assumption that $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$. If we can prove the $\frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}$ is a monotonic decreasing function for $x \in (0,1)$ and ranges from 0 to 1, then we can conclude that $f_{n_1,n_2,N_1,N_2,j}(x) \geq 0$ as desired.

Define $r(x) = r_{n_1,n_2,N_1,N_2,j}(x) = \frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}$, and notice that

$$r_{n_1,n_2,N_1,N_2,j}(x) = \left(\frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{a}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)}\right)^{b},$$

where $a = \frac{N_1 - N_2}{n_1 - n_2}$ and $b = n_1 - n_2$. The conditions of the proposition imply that $b \leq 0$ and $1 \leq a \leq \frac{N_2}{n_2}$. When $a = 1$ we have that $r(x) \in (0,1)$ since $I^{-1}_{N_2-j,j+1}(x) > I^{-1}_{n_2-j,j+1}(x)$.

Now let $s_a(x) = s_{n_2,N_2,a}(x) = \frac{I^{-1}_{N_2-j,j+1}(x)^a}{I^{-1}_{n_2-j,j+1}(x)}$. Then, when $a = \frac{N_2}{n_2}$, we have that

$$s_{\frac{N_2}{n_2}}(x) = \frac{I^{-1}_{N_2-j,j+1}(x)^{\frac{N_2}{n_2}}}{I^{-1}_{n_2-j,j+1}(x)} = \left(\frac{I^{-1}_{N_2-j,j+1}(x)^{N_2}}{I^{-1}_{n_2-j,j+1}(x)^{n_2}}\right)^{\frac{1}{n_2}}.$$

Lemma 5.1 states that for all $x \in (0,1)$, $I_{n-j,j+1}(x^{\frac{1}{n}}) \geq I_{n-j+1,j+1}(x^{\frac{1}{n+1}})$, and this implies that $I^{-1}_{N_2-j,j+1}(x)^{N_2} \geq I^{-1}_{n_2-j,j+1}(x)^{n_2}$. Thus, $s_{\frac{N_2}{n_2}}(x) \geq 1$ for all $x \in (0,1)$.

Need to make coherent

Since $I^{-1}_{x^{\frac{1}{n}}}(n-j, j+1) \in (0,1)$, then for any $a \in (1, \frac{N_2}{n_2})$, $s_a(x) \geq 1$, and this implies that $r(x) \in (0,1)$.

Then we show $\frac{\left(I^{-1}_{N_2-j,j+1}(x)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x)\right)^{n_1-n_2}}$ is a monotonic decreasing function for $x \in (0,1)$

Need to make coherent

We can write

$$\begin{aligned}
s_a'(x) &= \frac{a(I^{-1}_{N_2-j,j+1}(x))^{a-1} I^{-1}_{n_2-j,j+1}(x)/I'_{N_2-j,j+1}(I^{-1}_{N_2-j,j+1}(x)) - (I^{-1}_{N_2-j,j+1}(x))^a/I'_{n_2-j,j+1}(I^{-1}_{n_2-j,j+1}(x))}{I^{-1}_{n_2-j,j+1}(x)^2} \\
&= T_a(x)\left(aI^{-1}_{n_2-j,j+1}(x)I'_{n_2-j,j+1}(I^{-1}_{n_2-j,j+1}(x)) - I^{-1}_{N_2-j,j+1}(x)I'_{N_2-j,j+1}(I^{-1}_{N_2-j,j+1}(x))\right) \\
&= T_a(x)\left(a\frac{I^{-1}_{n_2-j,j+1}(x)^{n_2-j}(1-I^{-1}_{n_2-j,j+1}(x))^j}{B(n_2-j,j+1)} - \frac{I^{-1}_{N_2-j,j+1}(x)^{N_2-j}(1-I^{-1}_{N_2-j,j+1}(x))^j}{B(N_2-j,j+1)}\right),
\end{aligned}$$

where

$$T_a(x) = T_{a,n_2,N_2,j}(x) = \frac{I^{-1}_{N_2-j,j+1}(x)^{a-1}}{I^{-1}_{n_2-j,j+1}(x)^2 I'_{N_2-j,j+1}(I^{-1}_{N_2-j,j+1}(x)) I'_{n_2-j,j+1}(I^{-1}_{n_2-j,j+1}(x))}.$$

Then,

$$s'_{\frac{N_2}{n_2}}(x) = N_2 T_{\frac{N_2}{n_2}}(x)\left(\frac{I^{-1}_{n_2-j,j+1}(x)^{n_2-j}(1-I^{-1}_{n_2-j,j+1}(x))^j}{n_2 B(n_2-j,j+1)} - \frac{I^{-1}_{N_2-j,j+1}(x)^{N_2-j}(1-I^{-1}_{N_2-j,j+1}(x))^j}{N_2 B(N_2-j,j+1)}\right).$$

Now let

$$m(x) = m_{n_2,N_2,j}(x) = \frac{I^{-1}_{n_2-j,j+1}(x)^{n_2-j}(1-I^{-1}_{n_2-j,j+1}(x))^j}{n_2 B(n_2-j,j+1)} - \frac{I^{-1}_{N_2-j,j+1}(x)^{N_2-j}(1-I^{-1}_{N_2-j,j+1}(x))^j}{N_2 B(N_2-j,j+1)}.$$

We can write

$$\begin{aligned}
m'(x) &= \frac{1}{n_2}\left(\frac{(n_2-j)I^{-1}_{n_2-j,j+1}(x)^{n_2-j-1}}{I^{-1}_{n_2-j,j+1}(x)^{n_2-j-1}(1-I^{-1}_{n_2-j,j+1}(x))^j}(1-I^{-1}_{n_2-j,j+1}(x))^j\right.\\
&\qquad\left. - \frac{jI^{-1}_{n_2-j,j+1}(x)^{n_2-j}}{I^{-1}_{n_2-j,j+1}(x)^{n_2-j-1}(1-I^{-1}_{n_2-j,j+1}(x))^j}(1-I^{-1}_{n_2-j,j+1}(x))^{j-1}\right)\\
&\quad - \frac{1}{N_2}\left(\frac{(N_2-j)I^{-1}_{N_2-j,j+1}(x)^{N_2-j-1}}{I^{-1}_{N_2-j,j+1}(x)^{N_2-j-1}(1-I^{-1}_{N_2-j,j+1}(x))^j}(1-I^{-1}_{N_2-j,j+1}(x))^j\right.\\
&\qquad\left. - \frac{jI^{-1}_{N_2-j,j+1}(x)^{N_2-j}}{I^{-1}_{N_2-j,j+1}(x)^{N_2-j-1}(1-I^{-1}_{N_2-j,j+1}(x))^j}(1-I^{-1}_{N_2-j,j+1}(x))^{j-1}\right)\\
&= \frac{1}{n_2}\left((n_2-j) - \frac{jI^{-1}_{n_2-j,j+1}(x)}{1-I^{-1}_{n_2-j,j+1}(x)}\right) - \frac{1}{N_2}\left((N_2-j) - \frac{jI^{-1}_{N_2-j,j+1}(x)}{1-I^{-1}_{N_2-j,j+1}(x)}\right)\\
&= j\left(\frac{1}{N_2(1-I^{-1}_{N_2-j,j+1}(x))} - \frac{1}{n_2(1-I^{-1}_{n_2-j,j+1}(x))}\right).
\end{aligned}$$

We have that $m(0) = m(1) = 0$, and $m'(0) < 0$.

EDITED UP TO HERE

Also $\lim_{x\to 1} m'(x) > 0$ as $\frac{1}{1-I^{-1}_{N_2-j,j+1}(x)} > \frac{1}{1-I^{-1}_{n_2-j,j+1}(x)}$. In details,

for $\forall\epsilon_1 > 0$ and $\forall\epsilon_2 > 0$, we can find $x \in (0,1)$ that satisfies $I^{-1}_{N_2-j,j+1}(x) > I^{-1}_{n_2-j,j+1}(x) - \epsilon_1$ and $1 - x > \epsilon_2$. We choose the $\epsilon = min(\epsilon_1, \epsilon_2)$

Then we can find $x > I^{-1}_{N_2-j,j+1}(1 + \frac{\epsilon}{(n_2-N_2)/n_2})$,

$$\begin{aligned}
\frac{1}{N_2(1-I^{-1}_{N_2-j,j+1}(x))} - \frac{1}{n_2(1-I^{-1}_{n_2-j,j+1}(x))} &= \frac{n_2(1-I^{-1}_{n_2-j,j+1}(x)) - N_2(1-I^{-1}_{N_2-j,j+1}(x))}{n_2N_2(1-I^{-1}_{n_2-j,j+1}(x))(1-I^{-1}_{N_2-j,j+1}(x))}\\
&> \frac{(n_2-N_2)(1-I^{-1}_{N_2-j,j+1}(x)) + n_2\epsilon}{n_2N_2(1-I^{-1}_{n_2-j,j+1}(x))(1-I^{-1}_{N_2-j,j+1}(x))}\\
&> \frac{(n_2-N_2)(-\frac{\epsilon}{(n_2-N_2)/n_2}) + n_2\epsilon}{n_2N_2(1-I^{-1}_{n_2-j,j+1}(x))(1-I^{-1}_{N_2-j,j+1}(x))}\\
&> 0
\end{aligned}$$

Therefore, $\lim_{x\to 1} m'(x) > 0$.

Define

$$w_{n_2,N_2}(x) = \frac{1}{N_2(1-I^{-1}_{N_2-j,j+1}(x))} - \frac{1}{n_2(1-I^{-1}_{n_2-j,j+1}(x))} = \frac{n_2w_1x - N_2w_2x}{n_2N_2(1-I^{-1}_{n_2-j,j+1}(x))(1-I^{-1}_{N_2-j,j+1}(x))}$$

where $w_1(x) = 1 - I^{-1}_{n_2-j,j+1}(x)$, $w_2(x) = 1 - I^{-1}_{N_2-j,j+1}(x)$

Since $w_1(x) > w_2(x)$ and $w_2(x)$ decays slower than for $w_2(x)$ for $x \in (0,1)$. Also, we know $w_{n_2,N_2}(0) < 0$.

Thus, $m(x) < 0$ for $x \in (0,1)$.

Now we show $r(x)$ is a decreasing function that ranges from 0 to 1.

Let $x^* = x^*_{n_1,n_2,N_1,N_2,j}$ be such that $f'_{n_1,n_2,N_1,N_2,j}(x^*) = 0$, then

$$r(x^*) = \frac{\left(I^{-1}_{N_2-j,j+1}(x^*)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x^*)\right)^{n_1-n_2}} = \frac{B(n_2-j,j+1)/B(n_1-j,j+1)}{B(N_2-j,j+1)/B(N_1-j,j+1)} = c,$$

where $c = c_{n_1,n_2,N_1,N_2,j} \in (0,1)$, again from the assumption that $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$.

Then, based on the property of $r(x)$, x^* is the unique value that satisfies the $r(x^*) = c$.

We can conclude $f(x) \geq 0$ for all $x \in (0,1)$.

Is this needed: Let $x^* = x^*_{n_1,n_2,N_1,N_2,j}$ be such that $f'_{n_1,n_2,N_1,N_2,j}(x^*) = 0$, then

$$\frac{\left(I^{-1}_{N_2-j,j+1}(x^*)\right)^{N_1-N_2}}{\left(I^{-1}_{n_2-j,j+1}(x^*)\right)^{n_1-n_2}} = \frac{B(n_2-j,j+1)/B(n_1-j,j+1)}{B(N_2-j,j+1)/B(N_1-j,j+1)} = c,$$

where $c = c_{n_1,n_2,N_1,N_2,j} \in (0,1)$, again from the assumption that $\frac{n_1}{N_1} \leq \frac{n_2}{N_2}$.

□

Proposition 5.3. *Suppose that conditions of Theorem 3.4 where the system inclusion function is $1_i(X_{i,j} \geq X_{(N_i-n_i+1)})$, $i = 1,2$. Suppose that $n_2 \geq n_1 > 1$, $N_1 \geq N_2 > 1$, and fix $k = j$. Then condition (4) holds, and the conclusions of Theorem 3.4 follow.*

Proof. For ease of notation we will let $g_{n,i}(x) = I_{i,n-i+1}(x)$. The proof will use two induction arguments. For the base case let $n_1 = n_2$ and $N_1 = N_2$. With these specifications we have that

$$g_{n_1,n_1-j} \circ g^{-1}_{n_2,n_2-j}(x) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(x) = 0.$$

For the induction step, choose some $m \leq n_1 - 1$ and suppose that

$$g_{n_1-m,n_1-m-j} \circ g^{-1}_{n_2,n_2-j}(t) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(t) \geq 0, \tag{7}$$

for $t \in (0,1)$ where $n_2 \geq n_1$ and $N_1 \geq N_2$. An integration by parts calculation applied to regularized incomplete beta functions gives

$$I_{\alpha,\beta}(x) = I_{\alpha+1,\beta}(x) + \frac{x^\alpha (1-x)^\beta}{\alpha B(\alpha,\beta)}.$$

Applying the above to (7) yields

$$\begin{aligned}
&g_{n_1-m-1,n_1-m-1-j} \circ g^{-1}_{n_2,n_2-j}(t) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(t) \\
&\quad = g_{n_1-m,n_1-m-j} \circ g^{-1}_{n_2,n_2-j}(t) - g_{N_1,N_1-j} \circ g^{-1}_{N_2,N_2-j}(t) \\
&\qquad + \frac{\left[g^{-1}_{n_2,n_2-j}(t)\right]^{n_1-m-1-j} \left[1 - g^{-1}_{n_2,n_2-j}(t)\right]^{j+1}}{(n_1-m-1-j)B(n_1-m-1-j, j+1)} \\
&\quad \geq 0.
\end{aligned}$$

A similar argument holds for iterative increases to N_1. Thus, condition (4) holds under are assumed conditions and the conclusions of Theorem 3.4 follow. □

When $\frac{n_1}{N_1} \geq \frac{n_2}{N_2}$ and $k = j$, then
$g_{n_1,k} \circ g^{-1}_{n_2,j}(t) - g_{N_1,k} \circ g^{-1}_{N_2,j}(t)$ is non-negative when $t \in (0,1)$, where $g_{n,i}(x) = I_x(i, n-i+1)$, $I_x(\alpha,\beta)$ is the CDF of Beta(α,β) distribution.

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